LESSON 6.4

Write and Evaluate Numerical Expressions with Exponents

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can write numerical expressions from a real situation and evaluate them using the order of operations, including powers.

WHY THIS MATTERS The studio's booking sheet lists a price for each kind of seat. Writing one numerical expression for the whole order — and evaluating it in the right order — is faster and easier to check than adding costs one at a time. Some counts are written as powers, so the exponent has to be handled before any multiplying. The booking sheet lists a $150 base fee plus $12 for each speaker rented. Write one expression for the cost of renting 5 speakers.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Numerical expression *Spanish: Expresión numérica*	A math phrase built from numbers and operations, with no equal sign and no variable.	4 × 9 + 3 × 6
Evaluate *Spanish: Evaluar*	To find the single value an expression is worth by carrying out its operations in order.	3 × 6 = 18
Order of operations *Spanish: Orden de las operaciones*	The agreed order for evaluating: grouping symbols, then exponents, then multiply and divide left to right, then add and subtract left to right.	3 + 4 × 2^2 → 3 + 4 × 4 → 3 + 16 = 19
Power *Spanish: Potencia*	A short way to write repeated multiplication of the same factor, such as 2^3 for 2 × 2 × 2.	8 × 8 = 8^2
Base *Spanish: Base*	In a power, the number being multiplied by itself. In 2^3 the base is 2.	In 8^2, the base is 8.
Exponent *Spanish: Exponente*	In a power, the small raised number that says how many times the base is used as a factor.	In 8^2, the exponent is 2.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Ms. Ortiz buys 4 pizzas at $9 each and one $6 salad. Write an expression for the total. Evaluate it.

1. Pizzas are equal groups: 4 × 9.
2. Add the salad: 4 × 9 + 6.
3. Evaluate: 36 + 6 = 42.

Answer: 4 × 9 + 6 = $42

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Suppose the student tickets cost 2^3 dollars each instead of $2. The expression becomes $2 \times 15 + 4 \times 20 + 8 \times 2^3$.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The word _____ tells me to multiply _____ by _____ .
- The expression for this situation is _____ .
- I used a power because the same factor repeats _____ times.
- An expression has no equal sign, so I only _____ it.

Word bank numerical expression | evaluate | power | base | exponent | product | sum | difference | each | in all | operation | value

WATCH OUT!

A common mistake

A common mistake with powers is multiplying the base by the exponent: reading 3^2 as $3 \times 2 = 6$ instead of $3 \times 3 = 9$. A second is evaluating strictly left to right, so $3 + 4 \times 2$ becomes 14 instead of 11. The order of operations exists so one expression has exactly one value: grouping symbols, then powers, then × and ÷ left to right, then + and − left to right.

LESSON 6.4

Write and Evaluate Numerical Expressions with Exponents

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

What does 2^3 mean?

(A) 2 × 2 × 2
(B) 2 × 3
(C) 2 + 2 + 2
(D) 3 × 3

STEPS TO TRY
1. Which number is the base, and which is the exponent?
2. The exponent counts how many 2s are multiplied.
3. Write out the repeated multiplication the exponent asks for, then compare it with the choices.

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Sofia buys 4 tickets at $20 each and 2 tickets at $15 each. Which expression shows the total?

(A) 4 × 20 + 2 × 15

(B) 4 + 20 + 2 + 15

(C) (4 + 2) × (20 + 15)

(D) 4 × 2 + 20 × 15

STEPS TO TRY

1. What does "each" tell you to do?
2. Write one product per ticket type.
3. Then join the two costs with addition.

SHOW YOUR WORK

PART 2 Practice the skill

show every step

3 **Circle the letter of the best answer. Show how you know.**

Evaluate 3 + 4 × 2.

(A) 9

(B) 11

(C) 14

(D) 24

STEPS TO TRY

1. Which operation comes first in the order of operations?
2. Do 4 × 2 before touching the 3.
3. 3 + 8 = ?

SHOW EACH PARTIAL PRODUCT, THEN ADD THEM

1. **Split** *Break one factor apart.*
2. **Multiply** *One part at a time.*
3. **Add** *Add the parts.*

PART 3 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

Evaluate $5^2 + 3$.

(A) 13

(B) 16

(C) 28

(D) 64

STEPS TO TRY

1. What must you evaluate first?
2. $5^2 = 5 \times 5$.
3. $25 + 3 = ?$

Rewrite the whole expression each line	What I simplified

1. **Grouping** *Inside ( ) first.*
2. **Exponents** *Then powers.*
3. **× ÷** *Left to right.*
4. **+ −** *Left to right.*

EXPLAIN YOUR THINKING

Two students evaluate $3 + 4 \times 2$ and get 14 and 11. Which one followed the order of operations, and why does the order have to be agreed on at all?

SENTENCE STARTER The correct value is ___ because ___ comes before ___. Without an agreed order, the same expression would have ___ values.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help